TASK 5: E-Commerce Return Rate Investigation

1. Introduction

E-commerce businesses face significant challenges due to product returns, which increase operational costs and affect customer satisfaction. This project investigates the factors influencing product returns using data analysis techniques. The objective is to identify return patterns and provide business recommendations to reduce return rates.

2.Business Problem

E-commerce companies process millions of orders every day. However, product returns create major operational challenges and financial losses.

High return rates lead to:

- Increased logistics costs
- Warehouse burden
- Customer dissatisfaction
- Revenue leakage
- Inventory management issues

Business teams want answers:

i. ● Why are customers returning products?
ii. ● Are some categories more prone to returns?
iii. ● Does pricing affect returns?
iv. ● Are delivery delays influencing customer behavior?
v. ● Can statistical analysis uncover hidden causes?

3. Objectives

- Analyze e-commerce return data.
- Identify factors affecting product returns.
- Detect unusual customer and order behavior.
- Perform statistical testing.
- Segment customers based on return behavior.
- Provide business insights and recommendations.

4. Dataset Description

A synthetic e-commerce dataset containing 1000 customer orders and 10 features was used.

Features:

- Order_ID
- Customer_ID
- Order_Date
- Category
- Price
- Discount_Percent
- Delivery_Days
- Customer_Rating
- Region
- Returned

Each record represents one customer order and whether the product was returned.

5.Tasks to Perform

Phase 1: Data Understanding

Tasks Performed

- Loaded the dataset.
- Examined the first few records.

- Checked data dimensions.
- Verified data types.
- Identified missing values.
- Checked duplicate records.

Findings

- Total Orders: 1000
- Total Features: 10
- Missing Values: None
- Duplicate Records: None
- Dataset was clean and ready for analysis.

Phase 2: Descriptive Statistical Analysis

Analysis Performed

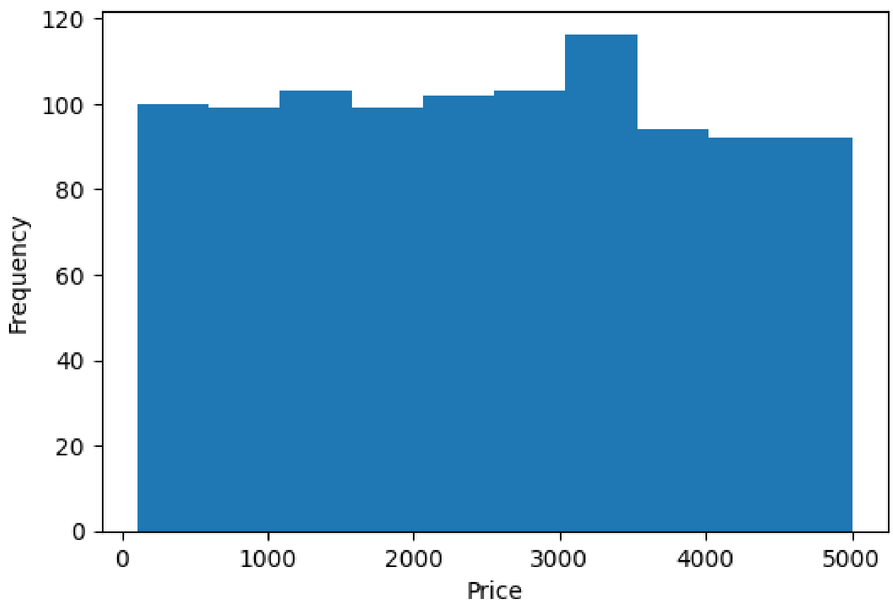
Descriptive statistics were calculated for:

- Product Price
- Delivery Days
- Customer Rating

Measures computed:

- Mean
- Median
- Standard Deviation
- Quartiles
- Distribution analysis

Histograms were used to understand data distributions



Findings

- Product prices varied across categories.
- Average delivery duration was approximately 8 days.
- Customer ratings ranged between 1 and 5.

Phase 3: Return Pattern Investigation

Analysis Performed

Relationships between returns and:

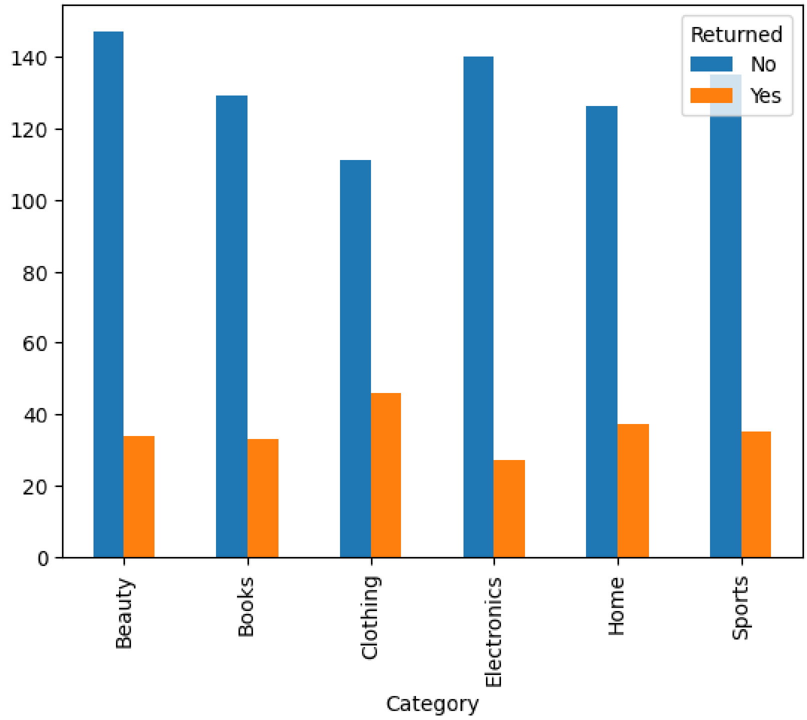
- Product category
- Product price
- Discount percentage
- Delivery duration
- Customer ratings

Bar charts and box plots were used.

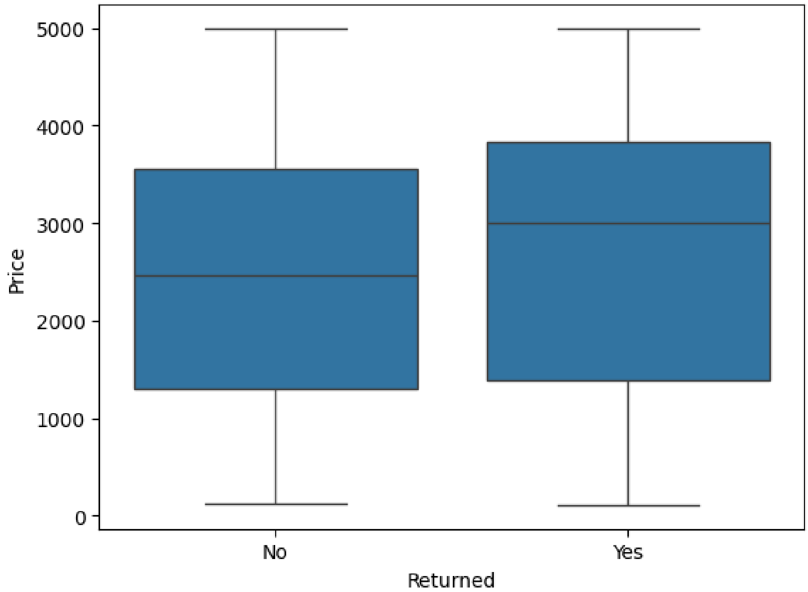
Findings

- Certain product categories had higher return rates.
- Higher discounts were associated with more returns.

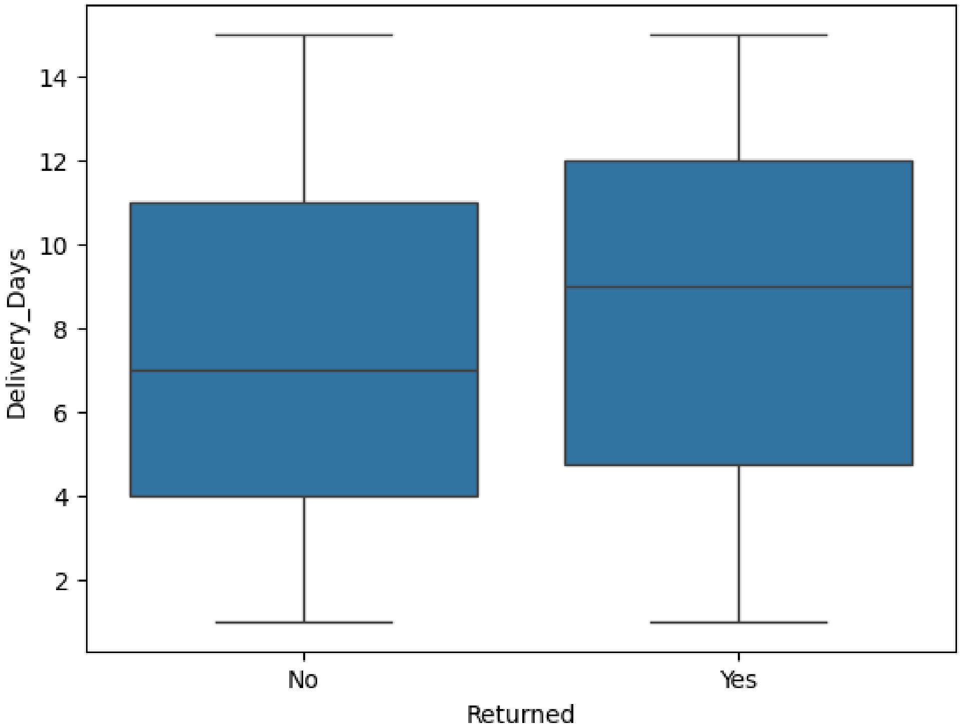
- Longer delivery times increased return probability.
- Lower customer ratings were linked with returns.



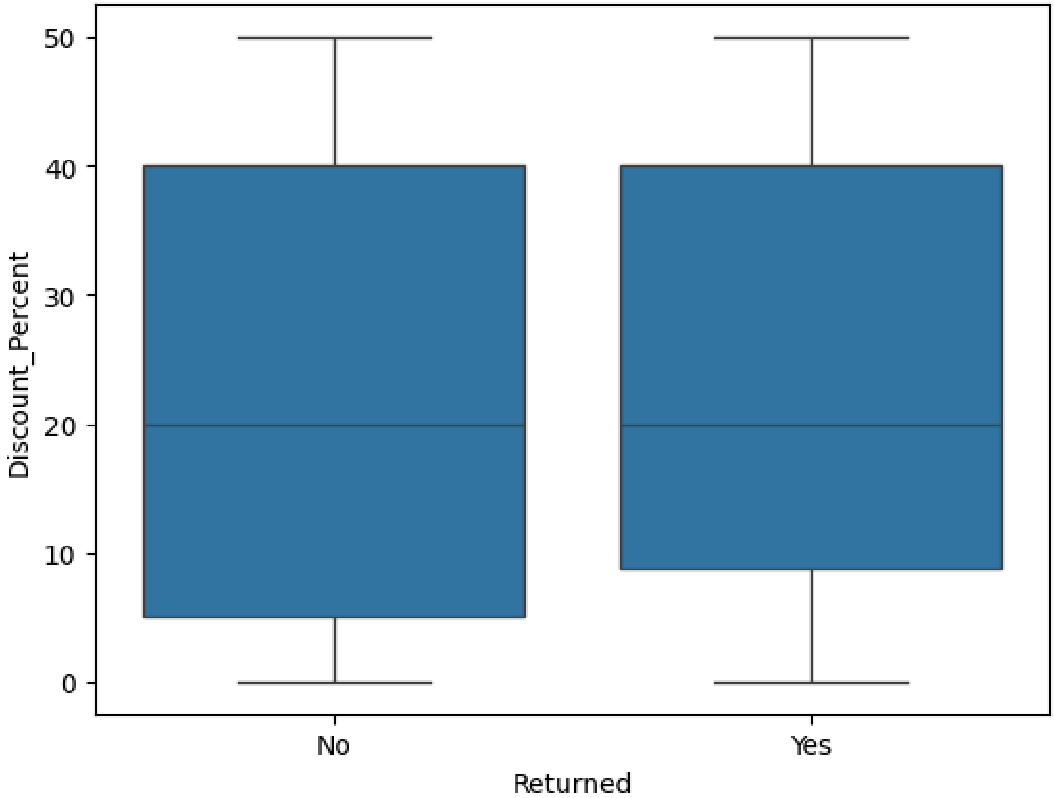
Category vs Return Rate



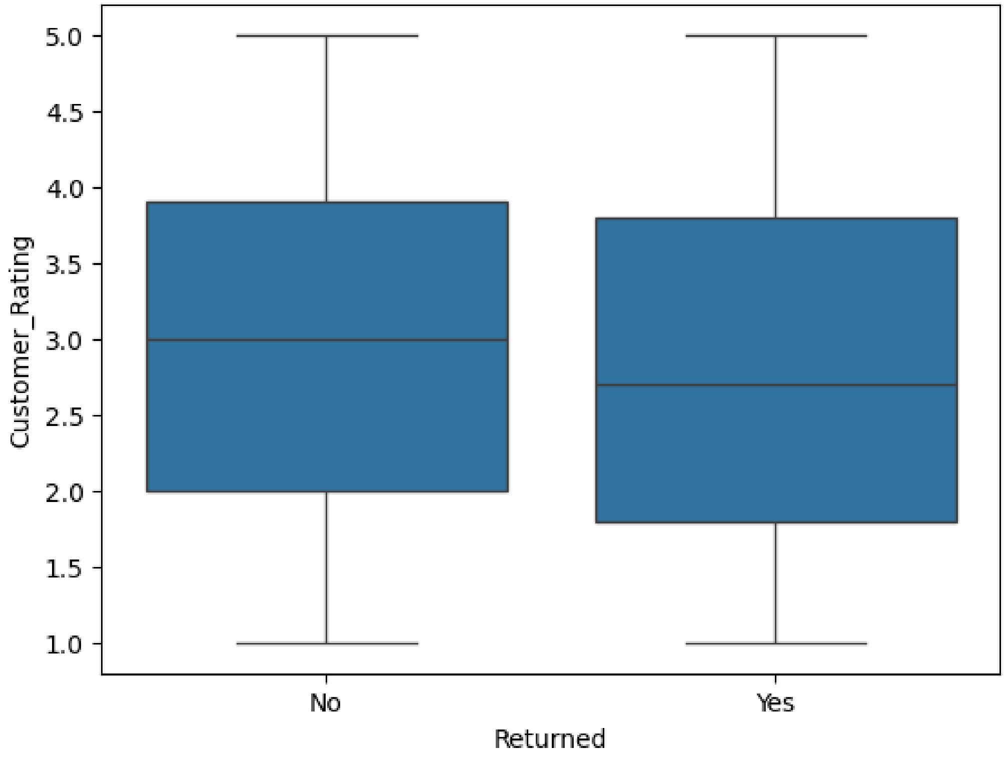
Price vs Return Rate



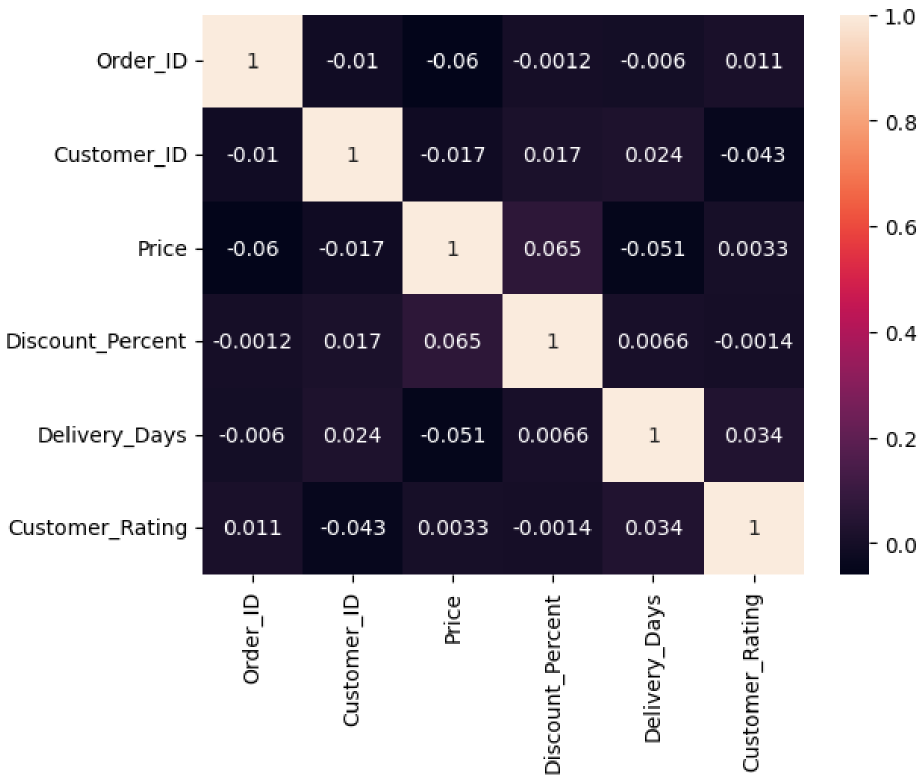
Delivery Time vs Return Rate



Discounts vs Return Rate

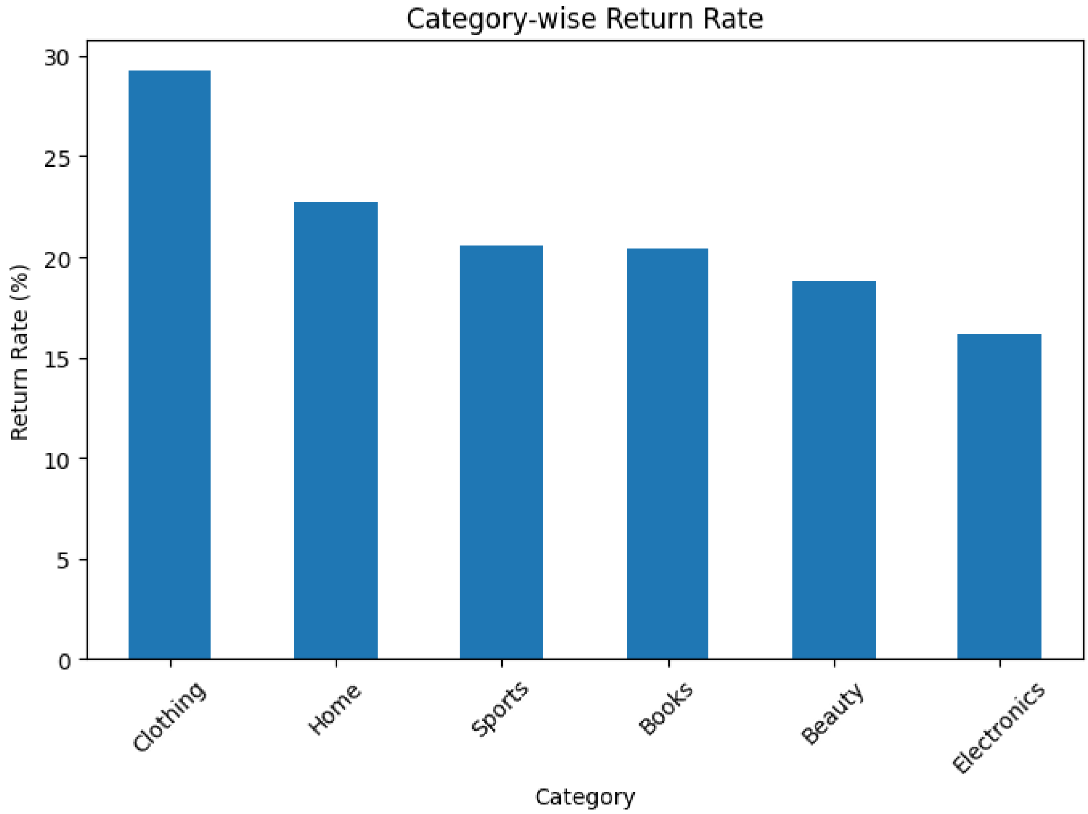


Customer Rating vs Return Rate

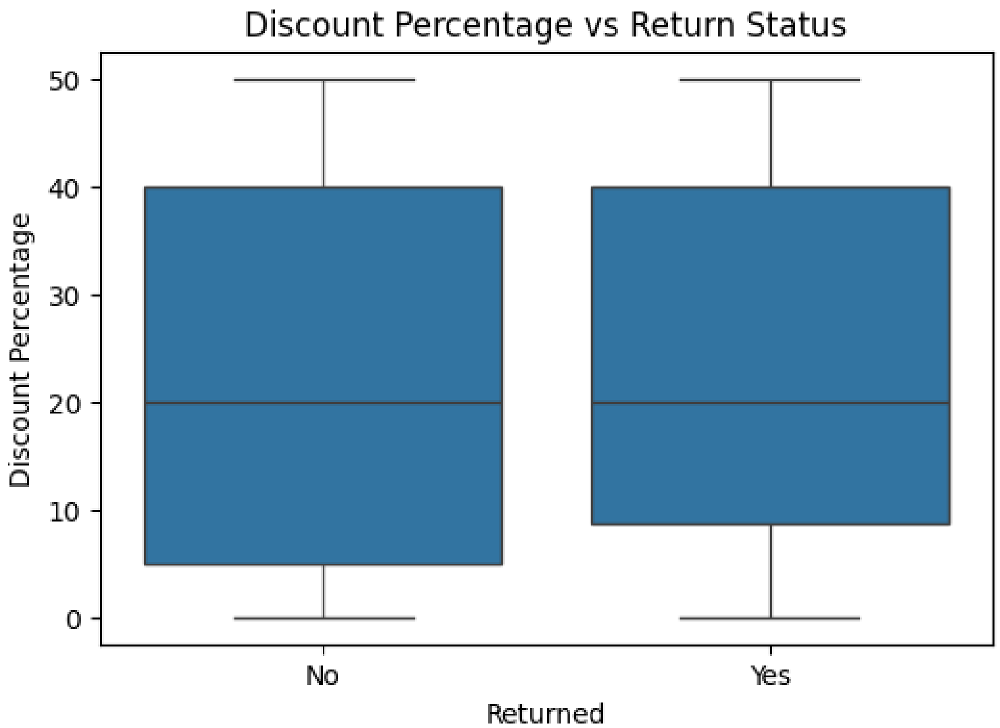


Heatmap

Which categories have highest return rates?



Does high discounting increase returns?



Phase 4: Outlier Analysis

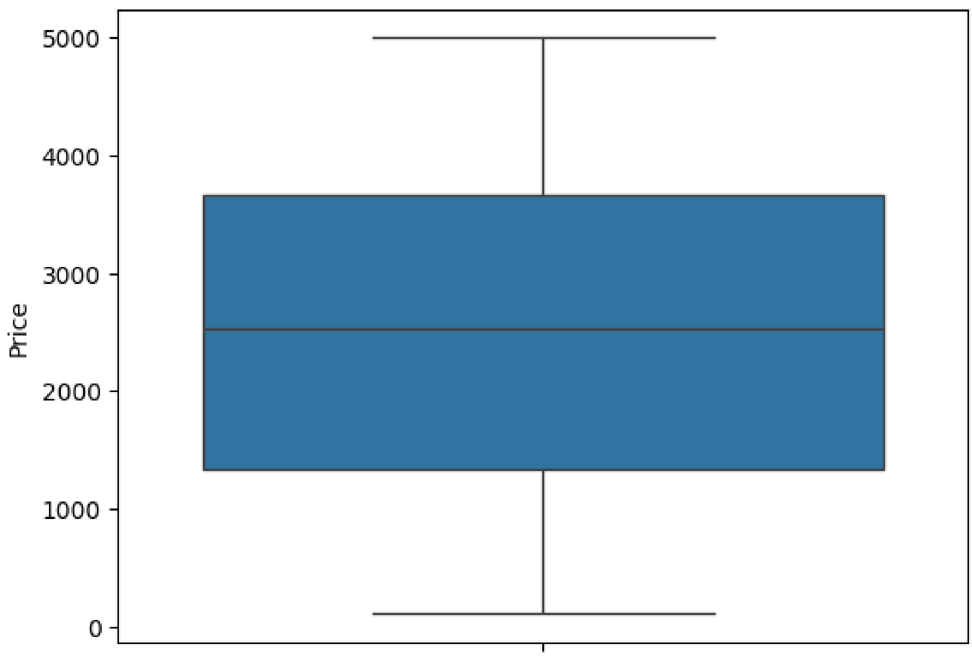
Methods

- Box plots
- Interquartile Range (IQR)
- Z-score method

Outliers Identified

- High-priced products.
- Customers with unusually high return frequencies.
- Orders with long delivery delays.

These observations may require business attention.



Box plots

Phase 5: Hypothesis Testing

Null Hypothesis (H0)

Delivery delay has no significant impact on product returns.

Alternative Hypothesis (H1)

Delivery delay significantly impacts product returns.

An independent sample t-test was performed.

Findings

Returned orders had longer average delivery times compared to non-returned orders.

Statistical testing suggested that delivery delays influence return behavior.

Phase 6: Customer Segmentation

Customers were grouped according to:

- Return frequency
- Purchase value
- Discount usage
- Delivery experience
- Customer ratings

Segments Identified

- High-return customers
- Premium customers
- Discount-driven customers
- Delayed delivery customers
- Low-rating customers

These segments help businesses design targeted strategies.

Phase 7: Business Insights

1. Which factors contribute most to returns?

The analysis indicates that delivery delays, higher discounts, and lower customer ratings are key factors influencing product returns. These factors increase the likelihood of customers returning purchased items.

2. Which categories are risky?

Among all product categories, Clothing has the highest return rate, followed by Home and Sports. These categories require closer monitoring and improvement strategies.

3. Does delayed delivery increase return probability?

Yes. The analysis suggests that longer delivery times are associated with a higher probability of product returns, highlighting the importance of efficient logistics.

4. Which customer groups need intervention?

High-return customers, discount-driven customers, and customers experiencing frequent delivery delays should be targeted for intervention to reduce return rates and improve satisfaction.

5. What actions should operations teams take?

Operations teams should focus on improving delivery efficiency, enhancing product quality and descriptions, optimizing discount policies, and monitoring high-return customer segments. These measures can help reduce unnecessary returns and improve overall business performance.

6.Conclusion

The E-commerce Return Rate Investigation helped identify the main factors affecting product returns. The analysis showed that delivery delays, discounts, customer ratings, and product categories influence return behavior. Clothing was found to have the highest return rate among all categories.

Customer segmentation and statistical analysis provided useful insights into customer behavior and return patterns. These findings can help e-commerce companies reduce return rates and improve customer satisfaction.

By improving delivery services, product quality, and return management strategies, businesses can reduce operational costs and increase overall efficiency.

Tools Used

- Google Colab
- Python
- Pandas
- NumPy
- Matplotlib
- SciPy

Techniques Used

- Data Cleaning
- Descriptive Statistics
- Data Visualization
- Return Pattern Analysis
- Outlier Detection
- Hypothesis Testing
- Customer Segmentation
- Business Analytics